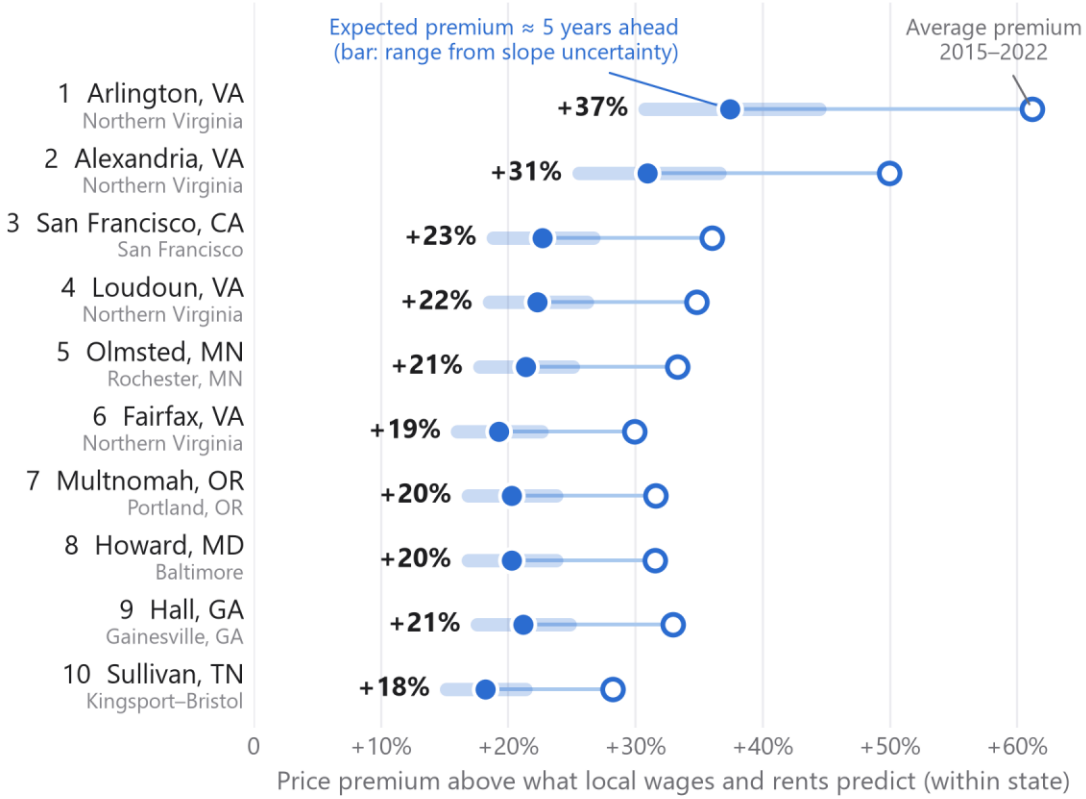
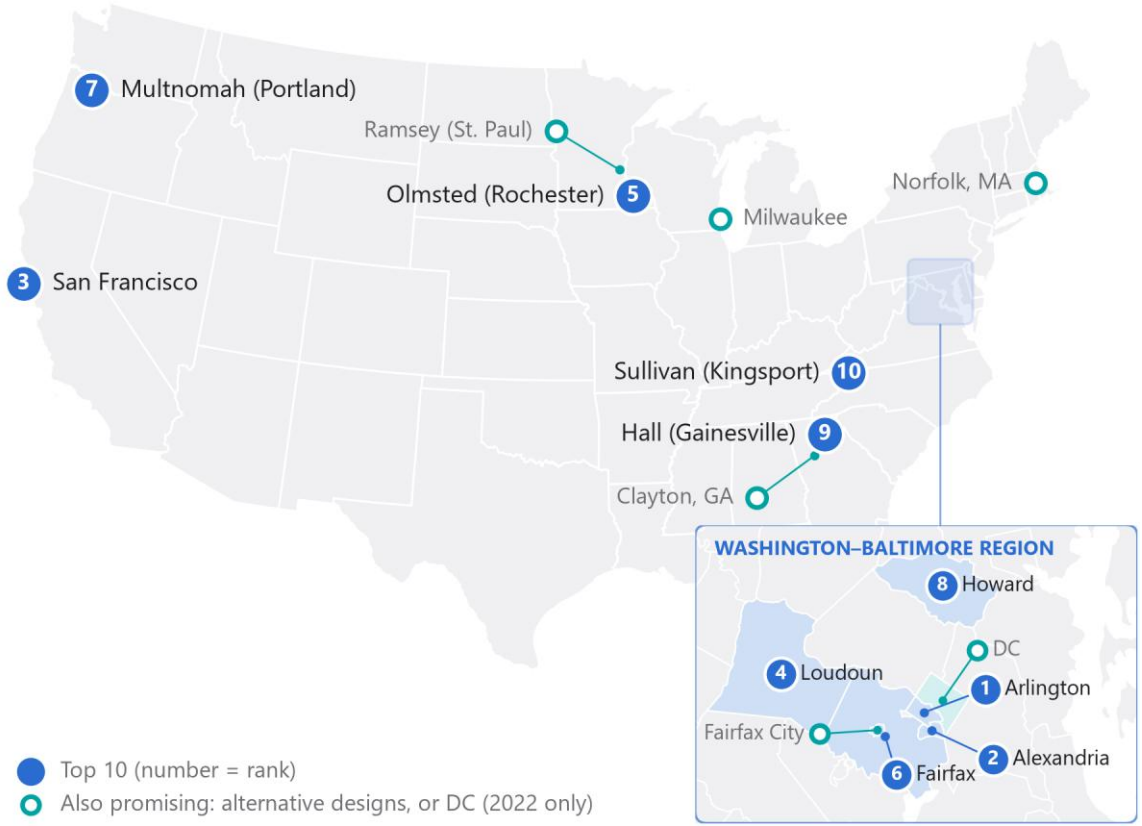
Ten counties where parents are expected to pay 18–37% more than local costs predict, an advantage that has lasted since 2015

TOP 10 · PRICE PREMIUM ABOVE WHAT LOCAL COSTS PREDICT



LOCATION · 1, 2, 4 AND 6 FORM ONE MARKET: NORTHERN VIRGINIA

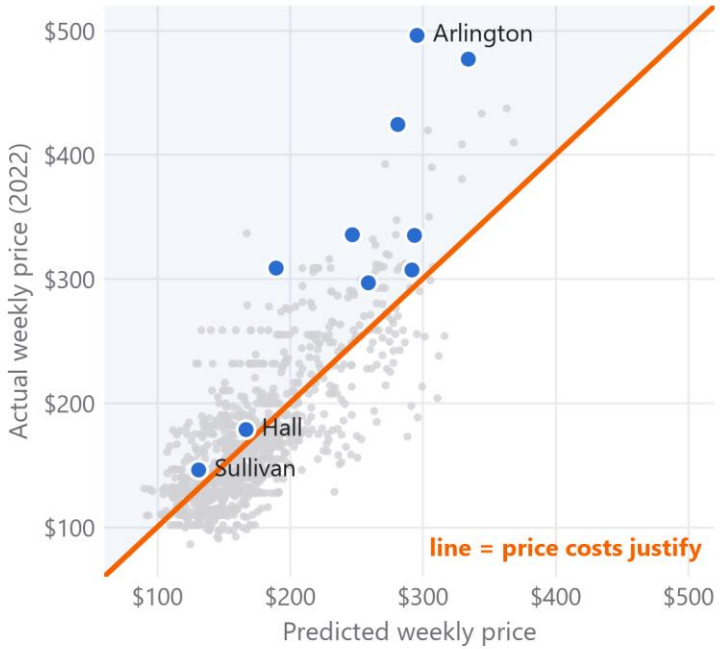


MGT 403 Data Project · Carson Tavares, Lee Itzhaki, Vusala Suleymanova, Philipp Romberg. Sample: 998 metro counties (2022), prices 2015–2022. Premium = actual price vs. price predicted from childcare-worker wages and rents within the same state; expected premium = after typical fading (slide 5); bar = range from the uncertainty of the fading slope.

Comparing each county with its own state lifts explained price variation from 61% to 81% and reveals hidden premiums

STEP 1 · MODEL "C"

Price = −88 + 7.09·wage + 5.91·rent



Above the line, parents pay more than costs justify. Here Hall and Sullivan look like ordinary markets.

STEP 2 · WITHIN STATE, IN %

+10% wage → +3.6% · +10% rent → +5.2%



Against their own state, all ten counties lie above the line.

WHY THE TWO STEPS

State dummies: states set rate zones and staffing rules, so each county is compared with its own state.
Logs: the gap becomes a % premium; $50 means more in Tennessee than in SF.

2022, N = 998	Model "c"	Final
Wage	7.09 (0.58)	0.37 (0.09)
Rent	5.91 (0.92)	0.53 (0.04)
State dummies	–	yes (43)
R^2	**0.61**	**0.81**

Model "c": $/week per $1,000 wage, $100 rent. Final: % change in price per 1% change. (SE, clustered by metro.) Like for like in $: R^2 0.61 → 0.80 with state effects; forecast error on unseen prices −14% (App. C).

EXAMPLE: HALL, GA (RANK 9), 2022

Parents pay **$179/week.**
Model "c" predicts **$167**: +7%, looks ordinary.
Within Georgia: **$138** predicted, so a **+30% premium.**

Model "c" (pre-analysis): price = a + b_1·wage + b_2·rent. Final: log(price) = a + b_1·log(wage) + b_2·log(rent) + state dummies, estimated per year 2015–2022. 2022: N = 998, 44 states/DC, SEs clustered by metro area. NDCP price, BLS OEWS childcare-worker wage, ACS median gross rent, 2022 $. Premium = residual. Appendix B–C.

From 1,159 counties to ten: a large, lasting price advantage, a market big enough to fill a center, and demand to rank them

THE FUNNEL: FROM ALL METRO COUNTIES TO TEN

1,159 — metro counties with price, wage and rent data
NDCP prices × BLS wages × ACS rents, 2015–2022

972 — in the 2022 data, observed ≥ 5 of 8 years
current market evidence + enough history

460 — positive premium also nationally
not just pricey relative to a cheap state

20 — largest average within-state premium
within-state model, averaged over 2015–2022

20 — premium positive in every single year
lasting, not a one-off spike

10 — ≥ 4,286 children needing care
median market size; the other 10 fail only here

Top 10 — ranked by score, removes none
50% premium + 50% demand rank

MARKET CHARACTERISTICS OF THE TEN

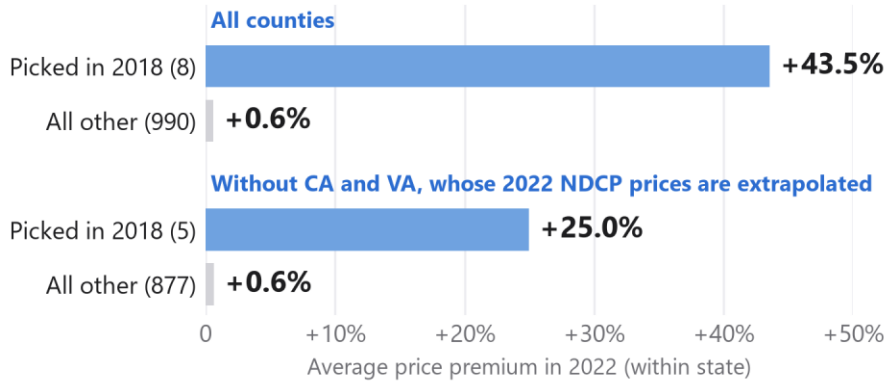
Color = percentile among all metro counties (0 – 100th)

		PRICE ADVANTAGE · 50%		DEMAND · 50%			
		Average premium 2015–2022	Weakest year	Children needing care	Care-need share	Income per capita	Score (0–1): premium + demand
1	Arlington, VA	+61% 100th pct	+47% 100th pct	11.6k 76th pct	4.9% 76th pct	$85k 100th pct	0.83
2	Alexandria, VA	+50% 100th pct	+35% 100th pct	8.8k 69th pct	5.6% 94th pct	$76k 99th pct	0.77
3	San Francisco, CA	+36% 100th pct	+21% 98th pct	30.0k 90th pct	3.5% 17th pct	$86k 100th pct	0.72
4	Loudoun, VA	+35% 100th pct	+31% 100th pct	22.3k 87th pct	5.3% 89th pct	$67k 99th pct	0.70
5	Olmsted, MN	+33% 99th pct	+32% 100th pct	9.5k 72th pct	5.9% 97th pct	$50k 90th pct	0.60
6	Fairfax, VA	+30% 99th pct	+26% 99th pct	57.3k 97th pct	5.0% 79th pct	$68k 99th pct	0.50
7	Multnomah, OR	+32% 99th pct	+22% 98th pct	32.2k 91th pct	4.0% 32th pct	$50k 90th pct	0.45
8	Howard, MD	+32% 99th pct	+25% 99th pct	16.3k 82th pct	4.9% 76th pct	$63k 98th pct	0.42
9	Hall, GA	+33% 99th pct	+28% 99th pct	9.2k 71th pct	4.5% 57th pct	$36k 46th pct	0.40
10	Sullivan, TN	+28% 98th pct	+24% 99th pct	5.6k 57th pct	3.5% 17th pct	$34k 38th pct	0.12

Premium = within-state % premium (slide 2), averaged over observed years 2015–2022 (≥ 5). Market size = children under 6 whose resident parents all work (ACS B23008). Score = 50% premium rank + 50% demand rank (children needing care, their share, income) among the ten; it only orders them. Female college share not used (r = 0.83 with income). Appendix D.

With 2018 data only, our rule already picked 7 of today's ten, and the list barely changes when we vary the design

BACKTEST: AS IF WE HAD DECIDED IN 2018



STATISTICAL CHECKS

≥ +27%

average premium of every one of the ten in its direct survey years only (no interpolated prices)

−14%

forecast error of our cost model vs. model "c" on 2022 prices it never saw (fitted 2015–20)

10/10

same list with both wage measures (8/10 if the teacher wage replaces ours)

ROBUSTNESS: 7 OF 10 SELECTED IN ALL 5 ALTERNATIVE DESIGNS

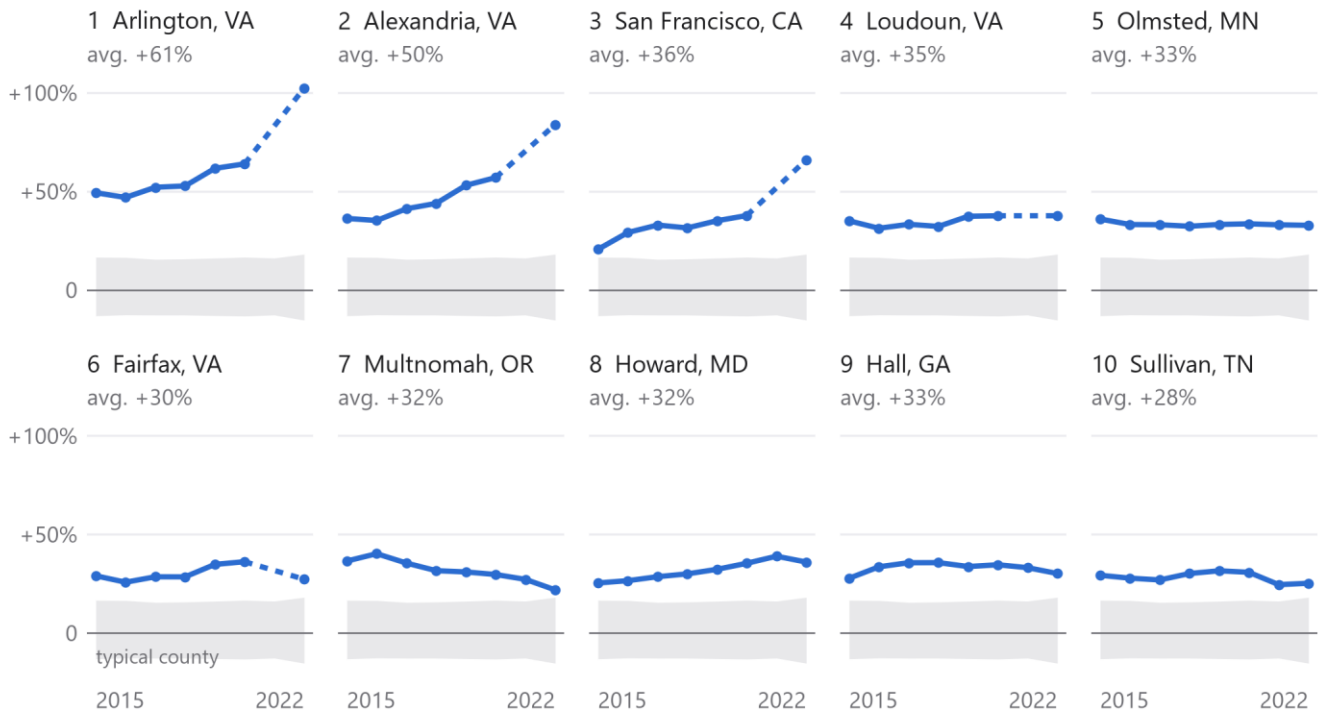
	Without 2022 data	Pre-COVID 2015–19	Premium in $ not %	Smaller markets	Top-30 pool	Kept
Arlington, VA	✓	✓	✓	✓	✓	5/5
Alexandria, VA	✓	✓	✓	✓	✓	5/5
San Francisco, CA	✓	✓	✓	✓	✓	5/5
Loudoun, VA	✓	✓	✓	✓	✓	5/5
Olmsted, MN	✓	✓	✓	✓	✓	5/5
Fairfax, VA	✓	✓	✓	✓	✓	5/5
Multnomah, OR	✓	✓	✓	✓	✓	5/5
Howard, MD	✓	–	–	✓	✓	3/5
Hall, GA	✓	✓	–	✓	✓	4/5
Sullivan, TN	–	✓	–	–	–	1/5

Not counted: weights 70/30 or 30/70 and "one weak year allowed" cannot change the list by construction. Weighting counties by children keeps 6 of 10.
Portfolio view (max. 2 per metro): Loudoun and Fairfax give way to Ramsey, MN and Trumbull, OH; the other 8 stay.

Backtest: complete rule applied with 2015–2018 data only (these years are also part of today's average); outcome = within-state premium in 2022 (8 of 10 picks still in the 2022 sample). Robustness: whole rule rerun with one change at a time. Appendix C and E.

Premiums above cost-based predictions last for years, but fade: about two thirds remain after five years

PREMIUM PER YEAR: EACH TOP-10 COUNTY VS. TYPICAL COUNTIES (GRAY: 10TH–90TH PCT.)



Dashed: 2021 missing (no NDCP prices for CA, VA). The 2022 jumps in Arlington, Alexandria and San Francisco come from NDCP-extrapolated CA/VA prices, so we rank on the 8-year average.

HOW MUCH OF A PREMIUM SURVIVES?

0.67

of a premium (log scale) remains ≈ 5 years later (survey prices only)

74%

of counties keep the sign of their premium

95% CI for the slope: [0.57, 0.78]; correlation 0.67

Counties in the top fifth in…	still top fifth in 2022	still positive
2015	56%	82%
2018	63%	89%
2019	73%	97%
If premiums were random	20%	≈ 50%

So we plan with the expected premium: each county's average premium, faded by the slope of 0.67 on the log scale: +18% to +37% for our ten (slide 1).

Within-state premium per county and year, 2015–2022 (log model, fitted per year). Survey test: two direct NDCP survey prices ≈ 4.7 years apart, 1,038 counties, 46 states, SE clustered by state. Quintile test: counties observed every year, incl. interpolated years (may overstate stability). Appendix D.

Our recommendation: open first in Northern Virginia, one market with four top counties, after checking local prices on the ground

FIVE LOCAL CHECKS THE COST MODEL CANNOT SEE

✓ typical ! watch ✕ risk (vs. all metro counties)

County	Competition	Child trend	Wage growth	Rent growth	Public pre-K*
Why it matters	Centers per 1,000 children needing care. ! above 6.24 (crowded) or below 3.86 (untested)	Children <6, 2015–22: future demand. ! below median (−4.4%), ✕ below −10.0%	2019–22: staff cost pressure. ! top quarter (above +6.1%)	2015–22: site cost pressure. ! top quarter (above +12.0%)	Free public pre-K competes directly. ! if free for all incomes (universal)
Arlington, VA	✓ 5.44 typical	✓ −3.3%	✓ +5.0%	✓ −1.3%	✓ targeted only
Alexandria, VA	✓ 5.67 typical	! −5.9%	✓ +5.0%	✓ +3.3%	✓ targeted only
San Francisco, CA	! 14.22 dense	! −5.5%	✓ −1.3%	! +20.4%	! universal (TK + city)
Loudoun, VA	! 6.54 dense	✓ +0.8%	✓ +5.0%	✓ +6.6%	✓ targeted only
Olmsted, MN	✓ 5.24 typical	! −5.7%	✓ −1.9%	! +16.8%	✓ targeted only
Fairfax, VA	✓ 6.22 typical	! −9.5%	✓ +5.0%	✓ +0.1%	✓ targeted only
Multnomah, OR	! 12.16 dense	✕ −15.8%	! +7.6%	! +27.3%	! Preschool for All
Howard, MD	! 7.00 dense	✓ +0.5%	! +8.6%	✓ −1.5%	✓ income-tiered only
Hall, GA	! 2.60 thin	! −5.1%	! +13.3%	✓ +11.3%	! universal (state)
Sullivan, TN	! 3.03 thin	! −6.4%	! +6.4%	✓ +6.7%	✓ targeted only

WHERE TO START: NORTHERN VIRGINIA

Arlington, Alexandria, Loudoun, Fairfax

Lasting premiums of +30% to +61%, large care-need markets, at most one minor flag.
One market, not four bets: shared wage, labor market and state survey, so open one site first. Virginia's last direct price surveys are from 2015 and 2018: check current prices locally.
Next in line: Olmsted and Howard.

NOT IN OUR DATA YET

- Commercial rents and sites (we use residential rent)
- Staff availability and turnover, not just average wages
- Local licensing rules and quality mix of competitors
- Remote work; federal job changes (DC area)
- Price data: only 2–5 direct survey years per county

Benchmarks: quartiles of all metro counties with data (882–974 counties). Competition: Census County Business Patterns (NAICS 624410, 2022), children needing care (ACS). *Public pre-K: state/county program rules, accessed Oct 2026 (sources in Appendix F).